

JAGAN SURENDRAN

Developer II - Software Engineering

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PROFESSIONAL SUMMARY

Experienced Software Engineer with expertise in Python, Flask, and FastAPI specializing in backend development, DevOps, application security, and AI-driven solutions. Skilled in building and maintaining scalable applications, integrating AI models (LLM, RAG, GenAI, Agentic AI), managing MySQL databases, manual testing, and optimizing CI/CD pipelines. Proficient in REST API development, API optimization, and secure authentication mechanisms. Adept at implementing AI-driven enhancements for dependency management tools, estimation tools, security analysis tools, and contract review applications. Led the development of intelligent applications including a vulnerability management platform and agent-based automation solutions to streamline workflows and improve decision-making using GenAI and Agentic AI.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, Node.js
 - **Frameworks & Libraries:** Flask, FastAPI, React, SQLAlchemy, LangChain, LangGraph, Pandas, Agno/Phidata
 - **AI & Machine Learning:** LLM (Large Language Models), RAG (Retrieval-Augmented Generation), GenAI (Generative AI), Agentic AI
 - **Databases:** MySQL, FAISS
 - **API Development:** REST API design, optimization, and data validation
 - **DevOps & Tools:** Azure Repos, Git, CI/CD, Docker
 - **Cloud Services:** Azure
 - **Security & Authentication:** API key-based authentication, Encryption mechanisms
 - **Other:** Dependency Management (POM, Gradle), Web Scraping, Testing & Validation
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WORK EXPERIENCE

Software Engineer – Developer II

UST | January 2022 – Present

- Designed and developed scalable backend services using Flask and FastAPI.
- Built and optimized REST APIs to ensure seamless communication between internal applications and external platforms.
- Integrated AI models (LLM, RAG, GenAI, Agentic AI) to enhance data retrieval, process automation, security, task execution, and decision-making.
- Developed agentic workflows for automating multi-step business processes, including security validation, contract reviews, and project estimation.
- Created modular AI agents capable of autonomously handling tasks such as data classification, anomaly detection, and project insights generation.
- Implemented secure authentication mechanisms, encryption techniques, and automated security validation using GenAI.
- Led the migration of UI components from HTML to React, improving user experience and maintainability.
- Developed tools for dependency validation and security analysis, ensuring software quality and integrity.

- Automated data synchronization and API integrations across multiple systems to improve operational efficiency
 - Contributed to DevOps processes, including CI/CD automation, sprint planning, and deployment strategies.
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PROJECTS & KEY CONTRIBUTIONS

- **Agentic AI-Powered Automation:** Developed intelligent agent-based systems for multi-step decision-making, project estimation, contract analysis, and data security workflows.
 - **AI-Driven Development:** Led the development of applications for secure coding, efficiency improvements, and intelligent decision-making using GenAI and Agentic AI, enhancing software security and business operations.
 - **AI-Driven Enhancements:** Optimized retrieval-based AI models and agentic planning for better search, classification and recommendation systems.
 - **Process Automation:** Developed AI-powered tools for user story generation, project estimation, and contract reviews.
 - **Security & Compliance:** Strengthened security through authentication, encryption, and automated security validation using GenAI.
 - **Visualization & Analysis:** Created tools for analysing market trends, project dependencies, and business objectives with LLM + agentic reasoning.
 - **Dependency & Risk Management:** Implemented automated dependency validation and security risk assessment processes.
 - **Manual Testing & QA:** Performed functional and usability testing for AI-powered applications, ensuring accuracy and stability.
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EDUCATION

BTech in Mechatronics

APJ Abdul Kalam Technological University | 2019

CERTIFICATIONS

- Oracle Cloud Infrastructure 2024 Generative AI Professional certification
 - Introduction to Prompt Engineering for Generative AI (2023)- LinkedIn certification
 - Certified Specialist in Data Science & Analytics - NORKA (2022)
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ADDITIONAL INFORMATION

- **Soft Skills:** Problem-solving, Adaptability, Continuous Learning, Time Management, Agile methodologies, Team collaboration
 - **Languages:** English, Malayalam
 - **Availability:** Open to new challenges and continuous learning
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